
Know When to Fold: Distribution-Shift-Aware Selective Prediction for Protein–Ligand Co-Folding

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Abstract

Protein–ligand co-folding models (AlphaFold3, Boltz, Chai) report confidence scores that correlate with pose accuracy, but a correlation is not a decision rule and it is weakest exactly where drug discovery operates, on novel pockets and novel chemotypes. We recast co-folding reliability as selective prediction and build a model-agnostic, training-free layer that turns confidence into a risk-controlled accept/abstain decision with a finite-sample guarantee. On the released Runs N’ Poses benchmark (five governed co-folding models, 12,602 delivered poses) a conformal selective-risk gate is valid i.i.d. yet breaks under training-similarity shift: a marginally compliant gate under-controls error about twofold on novel strata, and calibrating on familiar ligands then deploying on novel ones violates the guarantee in every run (realized error 0.55 against a 0.20 target). Group-conditional and weighted conformal keyed on training-set similarity repair per-stratum coverage, and a combined reliability score cuts the risk-coverage-curve area (AURC) by a quarter to two-fifths over native confidence, roughly doubling to tripling the predictions retained at a fixed guarantee. Structural pose-agreement features from the model’s own diffusion samples push the AURC reduction to 38–51%, and abstention raises a non-circular downstream metric, crystal-contact recovery (0.90 versus 0.72 on accepted versus rejected poses). The layer ships as a pip-installable package over frozen model outputs.

1 Introduction

Co-folding models are increasingly used to prioritise molecules and generate starting structures for downstream design. Recent work establishes that their confidence tracks accuracy and that accuracy drops on inputs dissimilar to training [7]. Neither gives a practitioner what they need: a rule for *which predictions to trust*, with a guarantee that survives the distribution shift they actually face.

We provide that rule. Our contribution is (i) selective prediction and conformal risk control for co-folding *pose* reliability, a structured 3D output rather than a scalar property. The closest prior work, CoDrug [3], conformalizes a scalar molecular property under shift, and we are not aware of a prior conformal treatment of pose correctness. Further, (ii) a sharp, quantified demonstration that standard conformal guarantees collapse under novel-pocket and novel-chemotype shift, using training-set structural and chemical similarity as the shift variable; and (iii) a shift-robust repair (group-conditional and importance-weighted conformal keyed on that similarity) plus a combined reliability score, packaged as a training-free layer over frozen AlphaFold3 (AF3), Boltz, and Chai outputs. We show ordinary probability calibration is not a substitute: it carries no finite-sample coverage and breaks under the same shift, whereas the conformal variants repair it.

2 Method

Setup. The unit is the delivered pose: the top-1 sample per (system, ligand, model) by the model’s own ranking score. The label is $Y = 1$ iff the symmetry-corrected ligand root-mean-square deviation

(RMSD) is at most 2 Å. The gate accepts a pose when its confidence $s \geq \tau$ and abstains otherwise; we report selective risk (error among accepted) against coverage (fraction accepted).

Certified threshold. We choose τ by Learn-then-Test with fixed-sequence testing [1] over a pre-specified, data-independent sequence of top- k accept ranks (smallest first). Each null $H_0(\tau)$: $P(\text{error} | s \geq \tau) \geq \alpha$ is tested with the binomial tail p-value $P(\text{Bin}(n_{\text{accept}}, \alpha) \leq \text{errors})$; the accept set grows while H_0 is rejected at level δ and stops at the first failure, giving $P(\text{selective risk} \leq \alpha) \geq 1 - \delta$, finite-sample and distribution-free. The p-value is exact for a homogeneous error rate and conservative, hence still valid, for the heterogeneous Poisson-binomial case by a convex-order argument; conditioning on the accept count is the correct tool for error-among-accepted.

Novelty and shift-robustness. Training-set similarity (ECFP4 Tanimoto to the nearest training ligand, pocket similarity, temporal cutoff), shipped pre-computed by Runs N’ Poses, is the covariate-shift variable. It both stratifies (group-conditional / Mondrian conformal [4]: a separate τ per novelty stratum) and weights (weighted conformal [2]: reweight familiar-source calibration points by an estimated likelihood ratio to certify on a novel target without target labels). Ligands with no training analog form their own extreme stratum.

Combined score. Native ranking score is a weak sorter for some models. We fit a calibration-only combiner (gradient boosting to $P(\text{correct})$) over native confidence, interface chain-pair ipTM, PoseBusters [9] physical validity, intra-model ensemble spread across diffusion samples, ligand difficulty, and cross-model agreement where other models’ predictions are available (a single-model deployment drops this feature, keeping the 22–38% gain without it). “Training-free” here means the co-folding model is never retrained; the combiner is a small model fit only on the calibration fold, so the layer is free of structure-predictor training, not of all fitting. Novelty is excluded from the score and enters only through calibration. A 3-way split (fit combiner / calibrate τ / test) preserves conformal validity for the learned score: the combiner never sees the test fold and the threshold is calibrated on out-of-sample combiner scores, so the learned score carries no combiner-fit leakage (target-clustering across folds is addressed under Utility).

3 Results

Data: released Runs N’ Poses predictions consumed as-is (no inference), five governed models (12,602 delivered poses); a Boltz-2 affinity head brings the raw table to 13,535 and is used only as an ungoverned comparator. $\alpha = 0.20$, $\delta = 0.10$ unless noted.

Raw novelty gradient. AF3 delivered-pose correctness falls from 0.88 on familiar ligands to 0.44 on the most novel with-analog stratum. A single global threshold cannot hold a uniform guarantee across this gradient.

Valid i.i.d. The certifier’s finite-sample guarantee, true selective risk $\leq \alpha$ with probability $1 - \delta$, is validated on synthetic data where the true risk is known (the gate holds in $\geq 1 - \delta$ of draws). The realized-risk indicator on a finite test fold is a downward-biased proxy for that event, and we demonstrate the bias rather than assert it: on synthetic data the small-fold indicator dips below $1 - \delta$ while the large-sample (true-risk) indicator meets it. On RNP the fraction of splits holding, with a Clopper-Pearson interval, reaches the $1 - \delta = 0.90$ target for Boltz-1x (0.90 [0.88, 0.93]) and approaches it for Boltz-1 (0.88 [0.85, 0.91]) and AF3 (0.85 [0.81, 0.89]), whose interval sits just below 0.90; the tight-certifier realized-risk indicator is the reason it is a downward-biased proxy rather than the guarantee itself, and the certified gate’s mean realized risk is $\leq \alpha$ there (AF3 0.17 at 28% coverage). Native ranking score certifies almost nothing for Chai and Protenix ($< 5\%$ coverage), a near-vacuous gate that motivates the combined score below.

The break. AF3’s marginal risk (0.177) hides severe per-stratum under-control (Table 1). Calibrating on familiar ligands and deploying on novel ones accepts 98% of poses at realized error 0.55 with the guarantee holding in 0% of runs. All five governed models show the same collapse. The break is a property of structural and chemical novelty, not recency: it is at least as strong along pocket-novelty (AF3 S_0 0.06 to S_3 0.40, Chai up to 0.77) but weak along a temporal release-date

Table 1: The exchangeability break: AF3 realized selective risk per novelty stratum (S_0 familiar to S_4 no analog) under a single global threshold. The marginal risk of 0.177 hides the tail.

novelty stratum	S_0	S_1	S_2	S_3	S_4
realized selective risk	0.07	0.15	0.16	0.38	0.43

Table 2: Area under the risk-coverage curve (AURC; lower is better), native versus combined score. The combined score cuts AURC by a quarter to two-fifths, with every per-model $\Delta(\text{AURC})$ 90% CI excluding zero (five models tested; Boltz-2 omitted for power).

model	AURC native	AURC combined	Δ
AF3	0.187	0.125	−33.1%
Boltz-1	0.234	0.170	−27.5%
Boltz-1x	0.216	0.163	−24.6%
Chai-1	0.248	0.150	−39.6%
Protenix	0.281	0.175	−37.7%

axis, which sharpens what the shift variable must capture.

The repair. Group-conditional calibration (a separate τ per novelty stratum) restores per-stratum validity: each stratum’s accepted error is $\leq \alpha$ with probability $1 - \delta$ marginally (a simultaneous across-strata statement follows by calibrating each at δ/S over the S strata). With native confidence the cost is heavy abstention on the hardest strata (the gate folds rather than assure falsely). With the combined score, group-conditional calibration instead recovers usable coverage across the gradient while holding risk $\leq \alpha$: AF3 accepts 52% of stratum S_1 and 39% of S_2 (against 8% and 12% with native confidence) and certifies a small fraction of the novel S_3 that native confidence must abstain on. Only the no-training-analog extreme (S_4) stays uncertifiable, where abstention is the correct answer.

Utility. The combined score dominates native confidence on the risk-coverage curve (Table 2). The AURC reduction is significant per model: a paired bootstrap over test poses gives 90% CIs on $\Delta(\text{AURC})$ that exclude zero (AF3 0.070 [0.049, 0.092], Protenix 0.092 [0.068, 0.116]; Boltz-2 omitted for power, $n = 933$). Because a target recurs across roughly five poses, the pooled random split leaks 95% of test targets into calibration; a target-grouped split and a target-cluster bootstrap give the same conclusion (AF3 $\Delta(\text{AURC})$ [0.056, 0.079]), so the gain is not an artifact of clustering. At $\alpha = 0.2$ the combined gate retains far more predictions at the same guarantee (AF3 coverage $0.22 \rightarrow 0.71$; Chai $0.01 \rightarrow 0.60$) and unlocks the stringent $\alpha = 0.1$ (90%-correct) operating point that native confidence cannot certify at all for several models.

Weighted repair, label-free. Calibrating on familiar ligands and correcting with cross-fitted, probability-calibrated likelihood-ratio weights over the novelty covariates pulls the realized error on a moderately-novel target toward α without target labels (AF3 naive 0.27 to weighted 0.19 at 0.70 coverage; same trend for all models, and robust to the weight-model choice). We also implement an importance-weighted Learn-then-Test gate [6] built on a WSR betting p-value [5] that is finite-sample valid conditional on correct weights; on real co-folding data it abstains, because even the plug-in barely clears α , so no certifiable margin remains. A concept-shift diagnostic explains why: $P(\text{correct} \mid \text{confidence})$ itself moves between source and target, and the gap grows from the moderate regime (0.08–0.16) to the extreme regime (0.17–0.29), so pure covariate reweighting cannot restore validity on novel pockets. Weighted conformal is therefore the label-free complement; group-conditional calibration is the rigorous finite-sample guarantee where stratum labels exist.

Baselines: calibration is not conformal. The combined score has the lowest AURC among the confidences a practitioner would reach for (AF3 0.12 versus native ipTM 0.16 versus ranking score 0.19; same ordering for all models), and PoseBusters validity alone is a poor gate (realized error

0.26–0.34). The decisive comparison isolates the guarantee from the features: the same combined score as a Platt/isotonic-calibrated classifier with a fixed threshold versus inside the conformal layer. Under i.i.d. data both control error; under the novelty shift the Platt-calibrated fixed-threshold gate breaks like naive conformal (realized error 0.22 above α). Isotonic calibration sits near α (0.18) only by abstaining down to 71% coverage, and even then holds the guarantee in 76% of runs rather than the required 90%, since calibration carries no finite-sample coverage. Only the shift-robust conformal repair (group-conditional, which has no calibration analogue) restores realized error $\leq \alpha$ with a guarantee, by abstaining more on the novel strata.

Downstream payoff. The gate cleans the delivered pose set a downstream pipeline would carry forward: base top-1 purity 63–70% rises to 82% ($\alpha = 0.2$, retaining 86% of correct AF3 poses) or 91–94% ($\alpha = 0.1$, all models), with interface LDDT-PLI lifting from about 0.72 to 0.85–0.92. Purity equals 1 minus selective risk by construction, so it cannot show the gate buys anything beyond the guarantee; we add a genuinely different, downstream metric: protein-ligand interaction-fingerprint recovery (receptor residues within 4.5 Å of the ligand versus the crystal structure), which is only correlated with the 2 Å label. Under the gate, accepted poses recover 0.90–0.91 of the crystal contacts versus 0.71–0.76 for rejected poses, and the accepted-minus-rejected recall gap 90% CI excludes zero for all five models (gaps 0.14–0.21). Abstention routes forward the poses whose interaction patterns a downstream step can trust. A screening-enrichment harness is implemented; the archival version applies it to a released co-folded screen [8], where the governed pose signal enriches actives above docking but the headline enrichment rides on an ungoverned affinity ranker, so we scope that arm out here.

Robustness, task, and label. The break and the gain do not depend on the 2 Å convention: across thresholds 1.5–3.0 Å the novel strata always exceed α (AF3 S_3 risk 0.35–0.43 versus S_0 0.05–0.09) and the combined score cuts AURC 31–34%. On interface quality (LDDT-PLI ≥ 0.5) the combined score again dominates native confidence on AURC (38–55% lower). Adding pose-agreement features from the model’s own 25 diffusion samples (intra-model diversity and cross-model delivered-pose RMSD, symmetry-corrected, no GPU) lowers AURC further, the single largest drop after ipTM (AF3 0.123 $\rightarrow$ 0.106), raising the overall reduction to 38–51% across models. Dropping the threshold and ordering by continuous RMSD, we certify a continuous gate (mean bounded-RMSD among accepted $\leq$ a target with probability $1 - \delta$) with a variance-adaptive WSR betting bound that certifies non-trivial coverage where a worst-case Hoeffding bound certifies almost none (AF3, 1 Å mean-RMSD target: 43% versus 0%).

4 Discussion and limitations

The guarantee is over the chosen correctness definition and the sampled shift axes, not a universal notion of trust; the findings are robust across RMSD thresholds 1.5–3.0 Å and extend to a certified continuous-RMSD gate. The weighted-conformal guarantee is exact only conditional on correct importance weights, and under novel-pocket shift there is residual concept shift, so we keep group-conditional as the rigorous finite-sample guarantee and scope weighted conformal as the label-free complement. A cross-dataset test on FoldBench recovers the withheld interface-ipTM by regenerating the Protenix predictions and self-scoring ligand-RMSD against the deposited assemblies (436 targets, 384 train-similar, 52 low-homology; feature and label from one run). This compares gates, not benchmarks: our self-scored Protenix top-1 success is 0.40 against FoldBench’s released 0.57, a gap from the scoring convention, model version, and MSA source that does not affect the transfer because feature and label are self-consistent. The frozen Runs N’ Poses interface-ipTM gate transfers with a risk-coverage AURC of 0.380, ahead of the matched ranking-score control at 0.454, and the same threshold accepts only 5% of FoldBench against 26% at home, so the coverage the gate can certify collapses on the low-homology second dataset. On the extreme novel-chemotype regime (base correctness $< 55\%$) no gate can certify a high-coverage low-error set; the layer’s value there is principled abstention over a naive gate’s false confidence. Code, calibration tables, and a one-command reproduction over the released predictions are provided in the accompanying repository (public release on acceptance).

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